

ESP32 S2 WROOM 1 IO ASSIGNMENT

https://docs.google.com/spreadsheets/d/1YJKf5_l8bu-Mrm7Ixr4pk0kDfURWdAnGm2sIA7eViG8/edit?gid=1051611186#gid=1051611186

Name	Pin No.	Type	Function	PCB assignment
GND	1	P	GND	GND
3V3	2	P	Power supply	3.3V
EN	3	I	High: on, enables the chip. Low: off, the chip powers off. Note: Do not leave the EN pin floating.	EN button
IO4	4	I/O/T	RTC_GPIO4, GPIO4, TOUCH4, ADC1_CH3	BATTERY ADC
IO5	5	I/O/T	RTC_GPIO5, GPIO5, TOUCH5, ADC1_CH4	
IO6	6	I/O/T	RTC_GPIO6, GPIO6, TOUCH6, ADC1_CH5	
IO7	7	I/O/T	RTC_GPIO7, GPIO7, TOUCH7, ADC1_CH6	
IO15	8	I/O/T	RTC_GPIO15, GPIO15, U0RTS, ADC2_CH4, XTAL_32K_P	GPS RESET
IO16	9	I/O/T	RTC_GPIO16, GPIO16, U0CTS, ADC2_CH5, XTAL_32K_N	GPS PPS
IO17	10	I/O/T	RTC_GPIO17, GPIO17, U1TXD, ADC2_CH6	GPS RX
IO18	11	I/O/T	RTC_GPIO18, GPIO18, U1RXD, ADC2_CH7, CLK_OUT3	GPS TX
IO8	12	I/O/T	RTC_GPIO8, GPIO8, TOUCH8, ADC1_CH7, SUBSPICS1	IMU SDA
IO19	13	I/O/T	RTC_GPIO19, GPIO19, U1RTS, ADC2_CH8, CLK_OUT2, USB_DI	
IO20	14	I/O/T	RTC_GPIO20, GPIO20, U1CTS, ADC2_CH9, CLK_OUT1, USB_D+	
IO3	15	I/O/T	RTC_GPIO3, GPIO3, TOUCH3, ADC1_CH2	
IO46	16	I/O/T	GPIO46	
IO9	17	I/O/T	RTC_GPIO9, GPIO9, TOUCH9, ADC1_CH8, FSPIHD, SUBSPIHD	IMU SCL
IO10	18	I/O/T	RTC_GPIO10, GPIO10, TOUCH10, ADC1_CH9, FSPICS0, FSPIIO4, SUBSPICS0	
IO11	19	I/O/T	RTC_GPIO11, GPIO11, TOUCH11, ADC2_CH0, FSPID, FSPIIO5, SUBSPID	
IO12	20	I/O/T	RTC_GPIO12, GPIO12, TOUCH12, ADC2_CH1, FSPICLK, FSPIIO6, SUBSPICLK	IMU INT
IO13	21	I/O/T	RTC_GPIO13, GPIO13, TOUCH13, ADC2_CH2, FSPIQ, FSPIIO7, SUBSPIQ	IMU RESET
IO14	22	I/O/T	RTC_GPIO14, GPIO14, TOUCH14, ADC2_CH3, FSPIWP, FSPIDQS, SUBSPIWP	GPS INT
IO21	23	I/O/T	RTC_GPIO21, GPIO21	
IO47	24	I/O/T	SPICLK_P_DIFF, GPIO47, SUBSPICLK_P_DIFF	
IO48	25	I/O/T	SPICLK_N_DIFF, GPIO48, SUBSPICLK_N_DIFF	
IO45	26	I/O/T	GPIO45	

IO0	27	I/O/T	RTC_GPIO0, GPIO0	Boot Button
IO35	28	I/O/T	SPIIO6, GPIO35, FSPID, SUBSPID	
IO36	29	I/O/T	SPIIO7, GPIO36, FSPICLK, SUBSPICLK	
IO37	30	I/O/T	SPIDQS, GPIO37, FSPIQ, SUBSPIQ	
IO38	31	I/O/T	GPIO38, FSPiWP, SUBSPiWP	Battery LED(RGB)
IO39	32	I/O/T	MTCK, GPIO39, CLK_OUT3, SUBSPICS1	Status LED(RGB)
IO40	33	I/O/T	MTDO, GPIO40, CLK_OUT2	WiFi LED
IO41	34	I/O/T	MTDI, GPIO41, CLK_OUT1	BLE LED
IO42	35	I/O/T	MTMS, GPIO42	
RXD0	36	I/O/T	U0RXD, GPIO44, CLK_OUT2	DEBUG UART TX
TXD0	37	I/O/T	U0TXD, GPIO43, CLK_OUT1	DEBUG UART RX
IO2	38	I/O/T	RTC_GPIO2, GPIO2, TOUCH2, ADC1_CH1	
IO1	39	I/O/T	RTC_GPIO1, GPIO1, TOUCH1, ADC1_CH0	
GND	40	P	GND	GND
EPAD	41	P	GND	GND